Dylan Becchetti

Mechanical Engineering | Manufacturing Automation & Engineering Software

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Portfolio: formsby-systems.github.io/engineering-portfolio

ENGINEERING EXPERIENCE

Cardinal Glass Industries | St. Louis Park, MN

Continuous Improvement Engineering Intern, summers 2024–2025 | Engineering Intern, summer 2023

- Built Excel/VBA reporting used in daily R&D debriefs. Automated weekly defect counts from 10,000+ unit records and linked production, yield and work orders.
- Linked purchase-order history to a legacy inventory in Smartsheet, reconciled inconsistent part identifiers, and surfaced items for maintenance review.
- Wrote illustrated assembly instructions for customer samples and planned spare-parts storage and rack signage across 14 locations.
- Measured cleanroom airflow, analyzed pilot-line data in Minitab and drafted layouts in AutoCAD.

SELECTED PROJECTS

Wilbur Powerhouse | Mechanical Design, year-long team capstone

- Built the full-scale OpenSCAD assembly and widened the ring gear to 79.5 inches. Worked with the team on carrier FEA, CNC fabrication and a bore clearance revision.

Midas Futures | Creator & Technical Lead

- Built a Python research pipeline with repeatable replay, checkpointed cloud campaigns and shared-account simulation. Matched 228 session decisions against a frozen reference.
- Built an unattended queue that ran 288 variants at two cost settings, reused the completed result on duplicate delivery and rejected an invalid request.

Operalith | Founder & Product Lead, independent product

- Built a React/Next.js workspace that links tasks, projects and inventory. Designed PostgreSQL saves to handle retries, stale edits and undo.

Formsby | Creator

- Built a Next.js and PostgreSQL housing search that queues runs, deduplicates listings and ranks results for the signed-in user’s latest run.

Scanned Leverage and Optimized Production | Creator

- Built a Python/Luau system for converting gameplay observations into configurable original prototypes. Reproduced two saved prototype variants exactly.

EDUCATION

Lehigh University | B.S. Mechanical Engineering, expected January 2027

Materials Science and Engineering minor | Coursework expected December 2026 | CSWA, 2024

TECHNICAL SKILLS

Python, JavaScript, React/Next.js, SQL/PostgreSQL, Git, automated testing, OpenSCAD, SOLIDWORKS, AutoCAD, Excel/VBA, Smartsheet, Minitab.